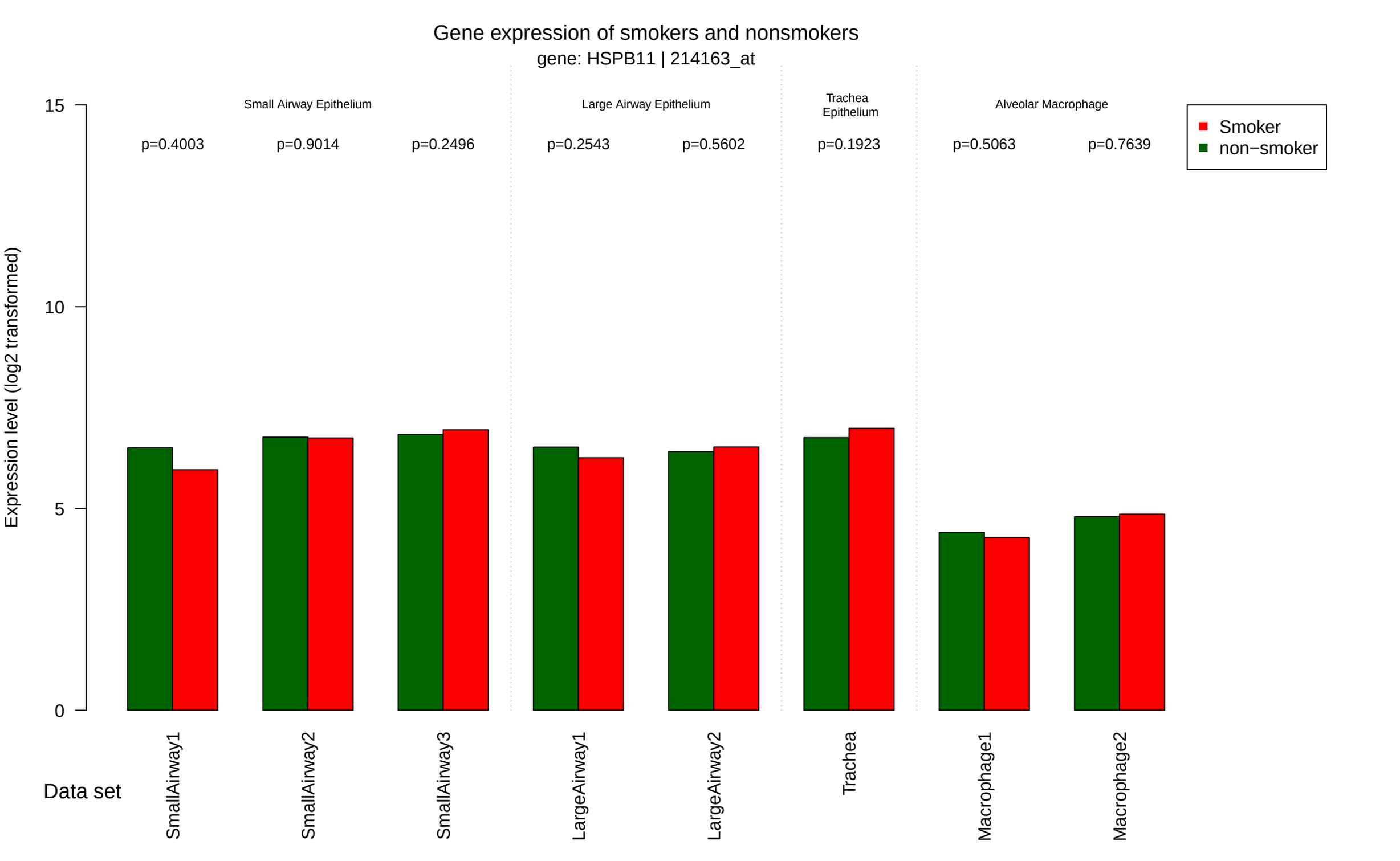
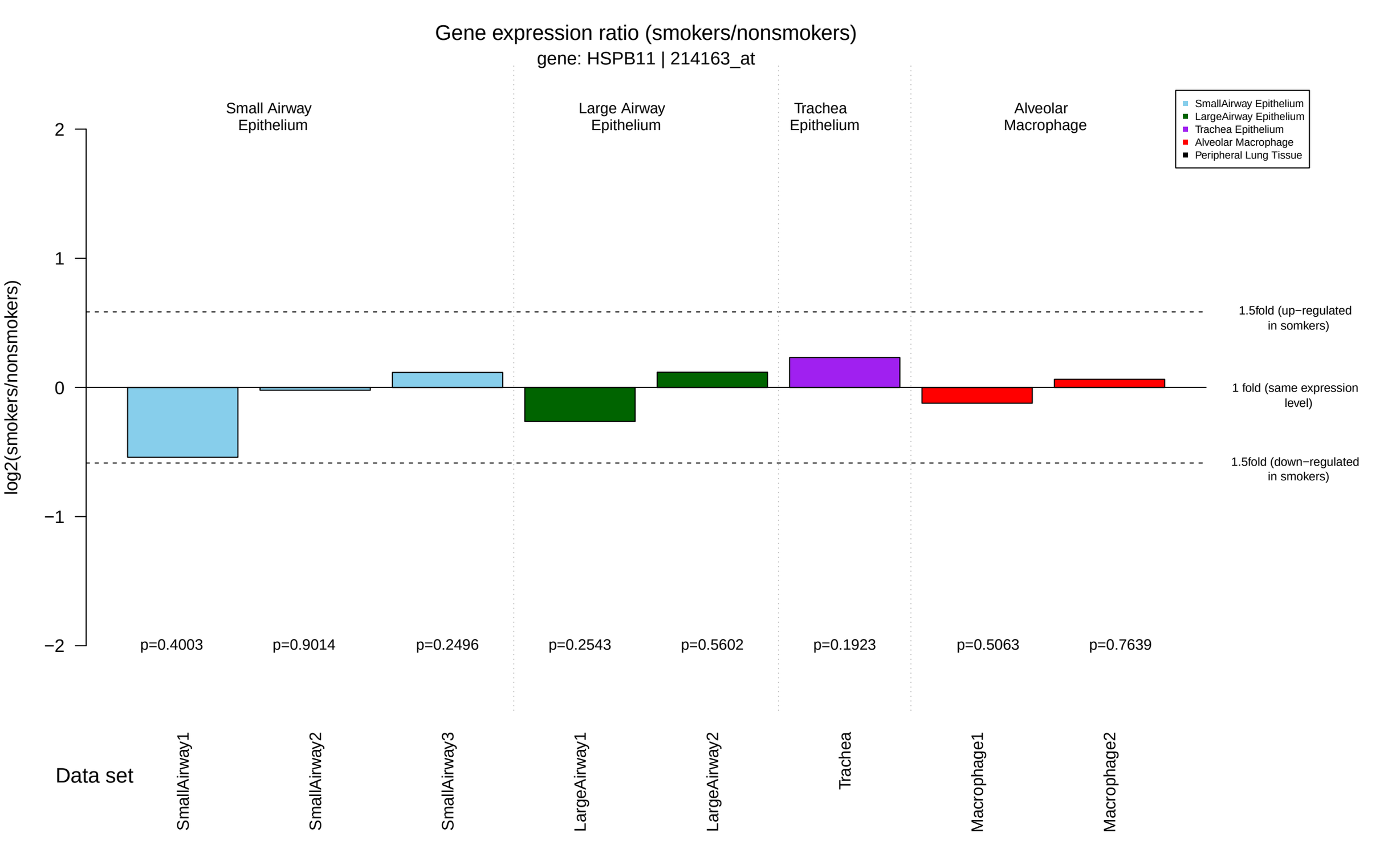
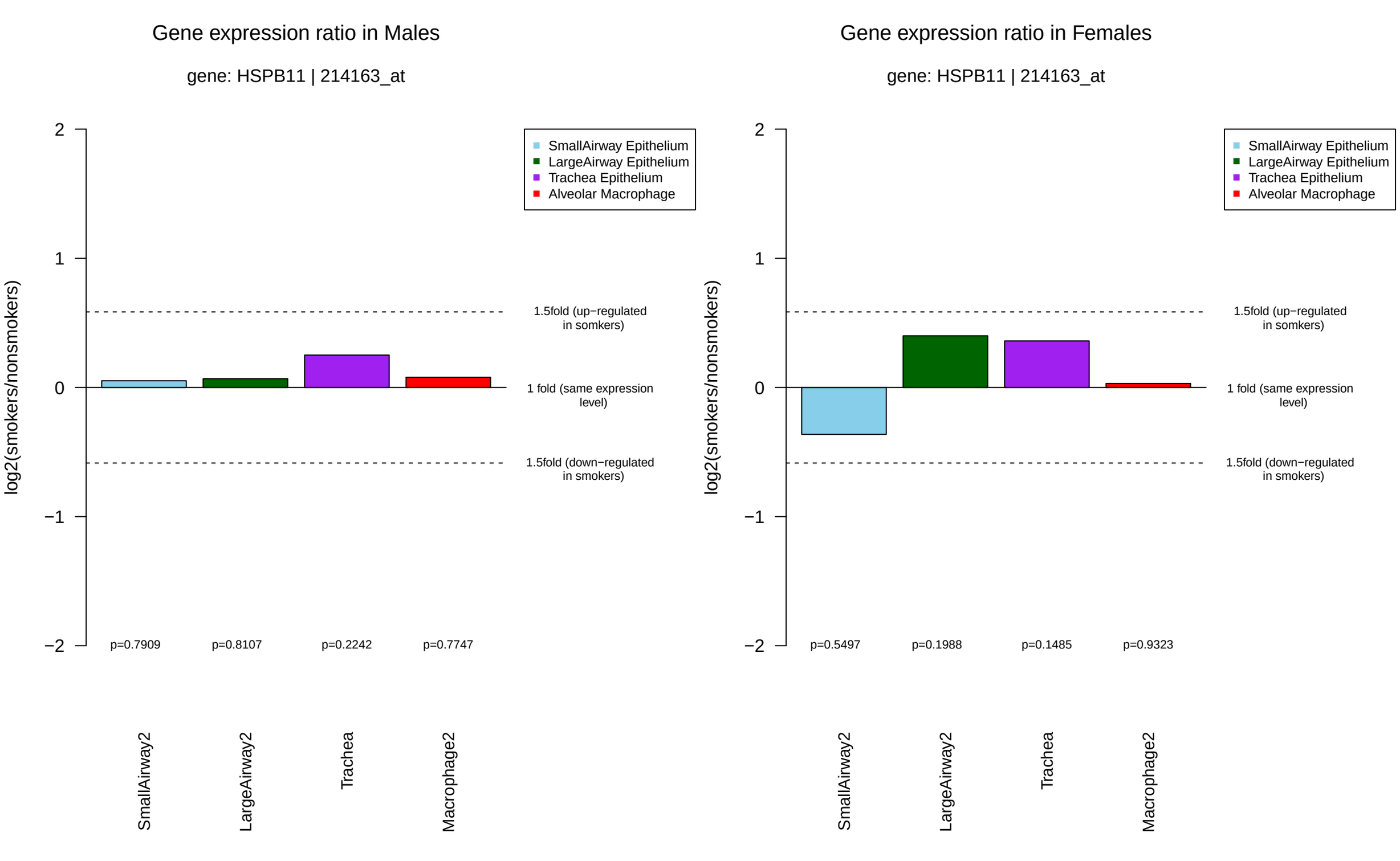
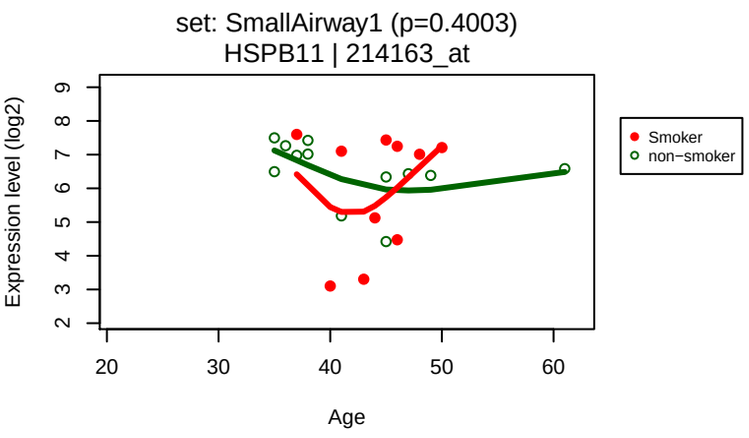
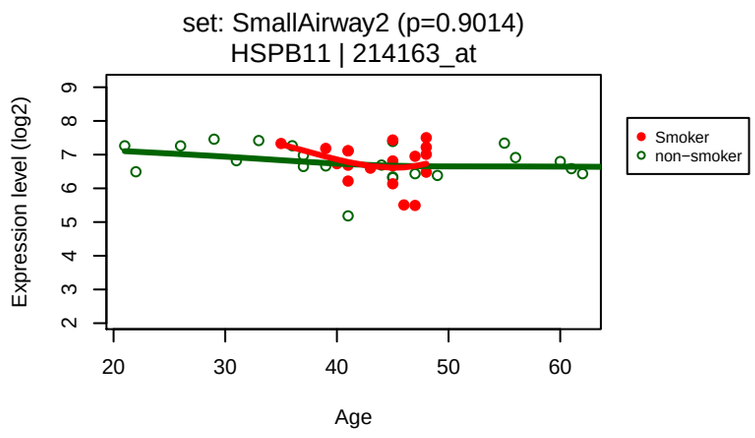
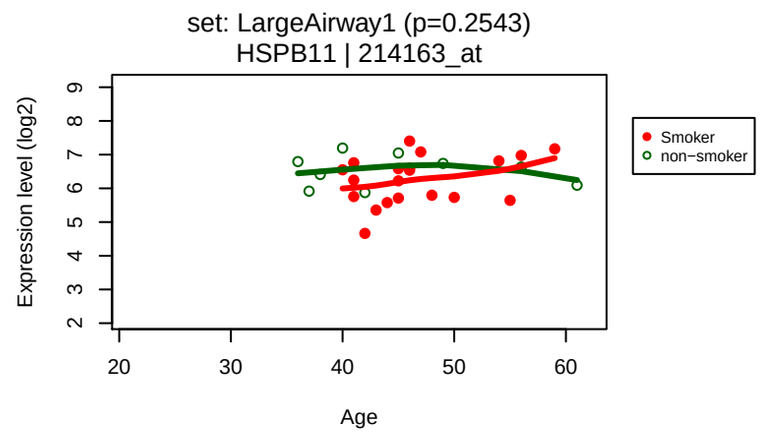

Gene expression of smokers and nonsmokers
gene: HSPB11 | 214163_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4003
p=0.9014
p=0.2496
p=0.2543
p=0.5602
p=0.1923
p=0.5063
p=0.7639
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

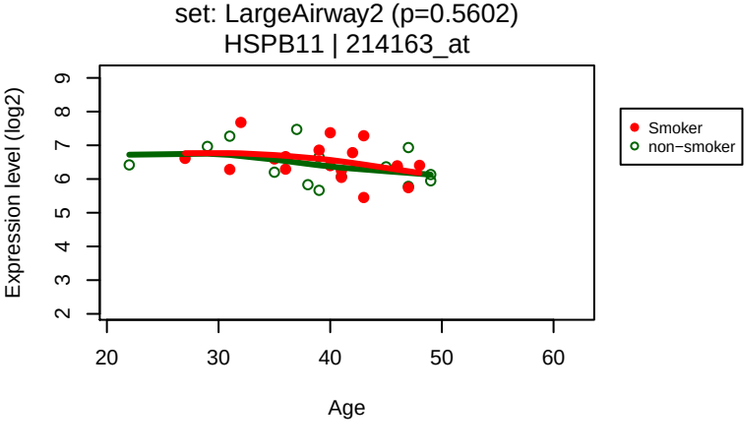

Gene expression ratio (smokers/nonsmokers)
gene: HSPB11 | 214163_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4003
p=0.9014
p=0.2496
p=0.2543
p=0.5602
p=0.1923
p=0.5063
p=0.7639
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

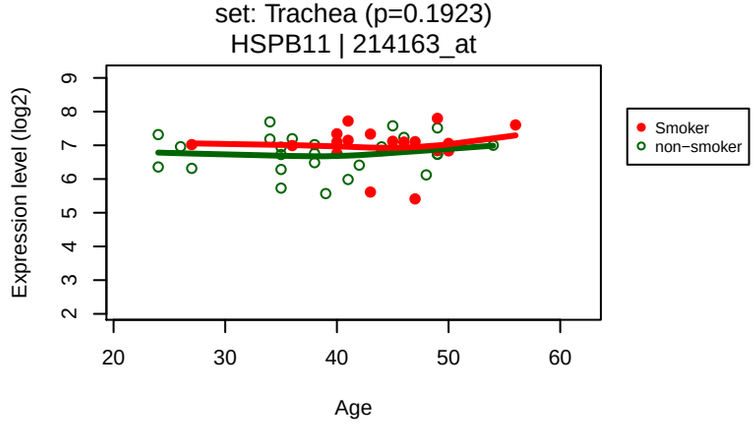

Gene expression ratio in Males
gene: HSPB11 | 214163_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7909
p=0.8107
p=0.2242
p=0.7747
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HSPB11 | 214163_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5497
p=0.1988
p=0.1485
p=0.9323
SmallAirway2
LargeAirway2
Trachea
Macrophage2

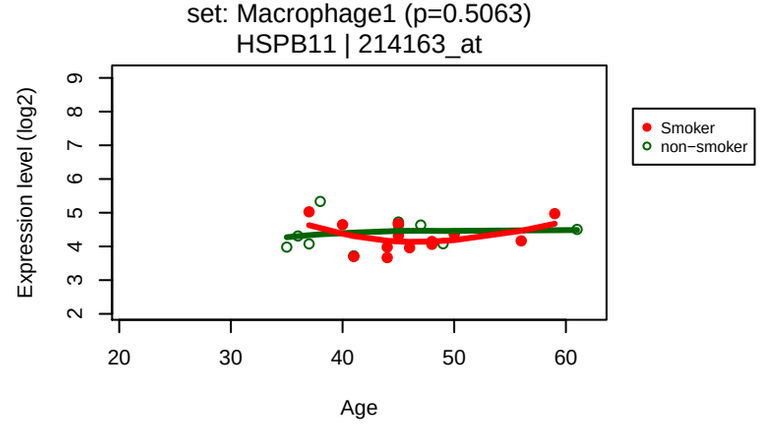

set: SmallAirway1 (p=0.4003)
HSPB11 | 214163_at
Expression level (log2)
Age
Smoker
non−smoker

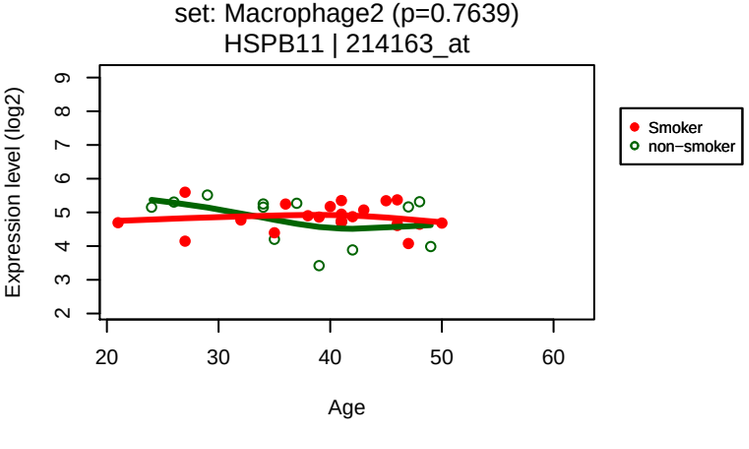

set: SmallAirway2 (p=0.9014)
HSPB11 | 214163_at
Expression level (log2)
Age
Smoker
non−smoker

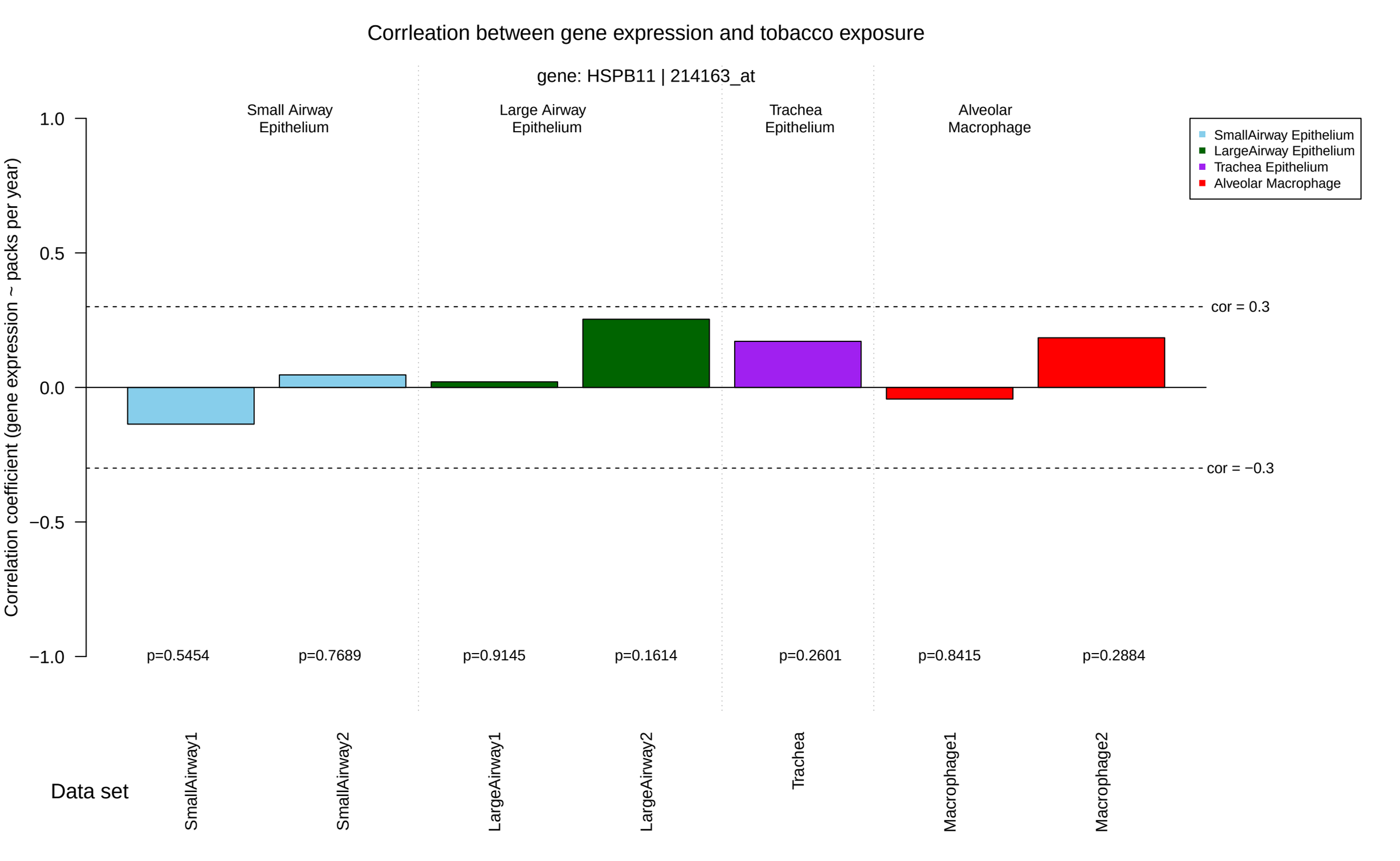

set: LargeAirway1 (p=0.2543)
HSPB11 | 214163_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5602)
HSPB11 | 214163_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1923)
HSPB11 | 214163_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5063)
HSPB11 | 214163_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7639)
HSPB11 | 214163_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HSPB11 | 214163_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5454
p=0.7689
p=0.9145
p=0.1614
p=0.2601
p=0.8415
p=0.2884
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2